Experiment 4 B

Aim

To extend the Library Book Issue Request Form in Zoho Creator using the Advanced Fields available in the Form Builder, enabling catalog verification, multi-book requests, fine computation, and automatic issue-ID/QR generation.

Fields Used

Field Type	Field Name	Sample Value / Behaviour
Decision Box	Requires_Home_Delivery	Yes / No
Rich Text	Library_Rules_Display	Formatted library rules shown inside the form
URL	Ebook_Reference_URL	https://library.rec.edu/ebooks/ai-fundamentals
Image	Book_Cover_Image	book_cover_ai_fundamentals.jpg
Percent	Late_Fee_Waiver_Percent	50%
Currency	Fine_Amount_Due	Rs. 45.00
Decimal	Book_Replacement_Price	650.50
Date-Time	Request_Submitted_On	24-Jul-2026 10:15
SubForm	Additional_Books_Requested	Introduction to Algorithms - Cormen - CSE
Lookup	Book_Title_Lookup	Autofills author, ISBN, shelf location from Book Catalog
Signature	Member_Digital_Signature	Signed (drawn signature)
File Upload	Faculty_Recommendation_Upload	faculty_recommendation.pdf
Audio	Special_Request_Voice_Note	voice_note_04.mp3 (0:12)
Video	Book_Review_Video	book_review.mp4 (0:35)
Auto Number	Issue_Request_ID	LIB-ISS-2026-000672
Formula	Total_Fine_Payable	Fine_Amount_Due - (Fine_Amount_Due x Late_Fee_Waiver_Percent)
QR/Barcode	Book_Issue_QR_Code	QR image encoding the Issue Request ID

Form Layout (Section-wise)

The sections below are added to the existing Member, Contact, and Book Request Details sections of the basic Library Book Issue Request Form.

Section E — Catalog Verification & Reference

- Book Title Lookup (Lookup field, auto-fills from master Book Catalog table)
- Book Cover Image (Image field)
- Ebook Reference URL (URL field)
- Book Replacement Price (Decimal field)

Section F — Multi-Book Request & Media Details

- Additional Books Requested (SubForm field, repeating rows)
- Faculty Recommendation Upload (File Upload field)
- Book Review Video (Video field, optional)

- Special Request Voice Note (Audio field, optional)

Section G — Fine & Delivery Preference

- Requires Home Delivery (Decision Box)
- Fine Amount Due (Currency field)
- Late Fee Waiver Percent (Percent field)
- Total Fine Payable (Formula field, read-only, auto-computed)

Section H — Declaration & System-Generated Values

- Library Rules Display (Rich Text field, read-only reference block)
- Member Digital Signature (Signature field)
- Request Submitted On (Date-Time field, auto-captured)
- Issue Request ID (Auto Number field, system-generated)
- Book Issue QR Code (QR/Barcode field, system-generated from Issue Request ID)

Validation & Configuration Rules

- Lookup field (Book Title Lookup): connected to a master Book Catalog form; members type-ahead search and select instead of typing free text.
- SubForm (Additional Books Requested): minimum 0 and maximum 4 rows per request.
- File Upload (Faculty Recommendation Upload): accept PDF, JPG, PNG only; maximum 5 MB.
- Image (Book Cover Image): accept JPG and PNG only; maximum 2 MB.
- Video (Book Review Video): maximum duration 45 seconds and maximum size 50 MB.
- Percent (Late Fee Waiver Percent): value range 0 to 100, editable only by library staff.
- Currency (Fine Amount Due): non-negative values only, formatted in INR.
- Formula (Total Fine Payable): read-only; recalculated whenever Fine Amount Due or Late Fee Waiver Percent changes.
- Auto Number (Issue Request ID): prefix "LIB-ISS-", the current year, and a 6-digit auto-incrementing sequence.
- QR/Barcode (Book Issue QR Code): auto-generated immediately after the Issue Request ID is assigned on submission.
- Signature (Member Digital Signature) and Decision Box (Requires Home Delivery) are mandatory; the form cannot be submitted without them.

Workflow Summary

- Step 1: Member completes the existing basic-field sections (personal, contact, and book category details).
- Step 2: Member selects the exact book using the Book Title Lookup field, which auto-fills the author, ISBN, and shelf location.
- Step 3: Additional books needed in the same visit are added as rows in the Additional Books Requested SubForm.
- Step 4: Member uploads a faculty recommendation (if required) and optionally a book review video or voice note.
- Step 5: Member indicates whether home delivery is required; Fine Amount Due, Late Fee Waiver and computed Total Fine Payable are displayed.

- Step 6: Member reviews the Library Rules block and provides a Digital Signature.
- Step 7: On Submit, Zoho Creator auto-generates the Issue Request ID, timestamps Request Submitted On, and generates the Book Issue QR Code.
- Step 8: Library staff review the Book_Issue_Request_Report, verify documents, confirm the fine against Total Fine Payable, and scan the QR code at the counter.

Screenshots

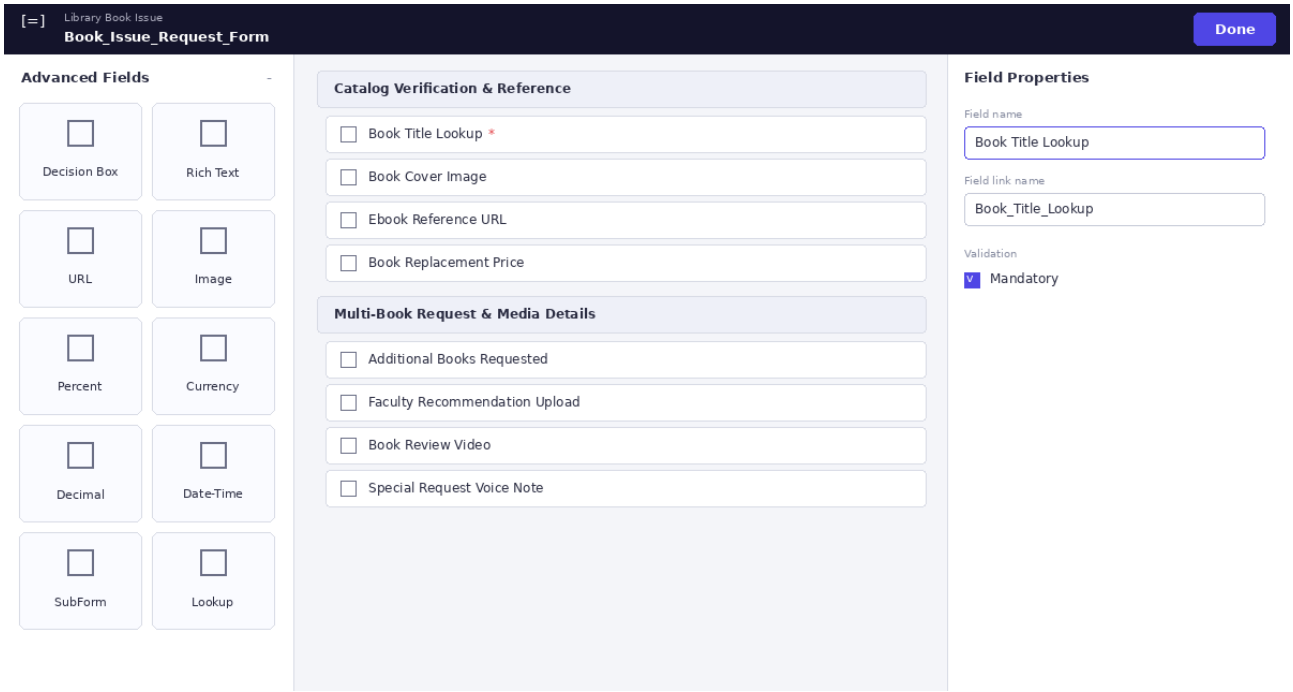


Figure 4B-1: Form Builder - Sections E and F, Advanced Fields (sample illustration)

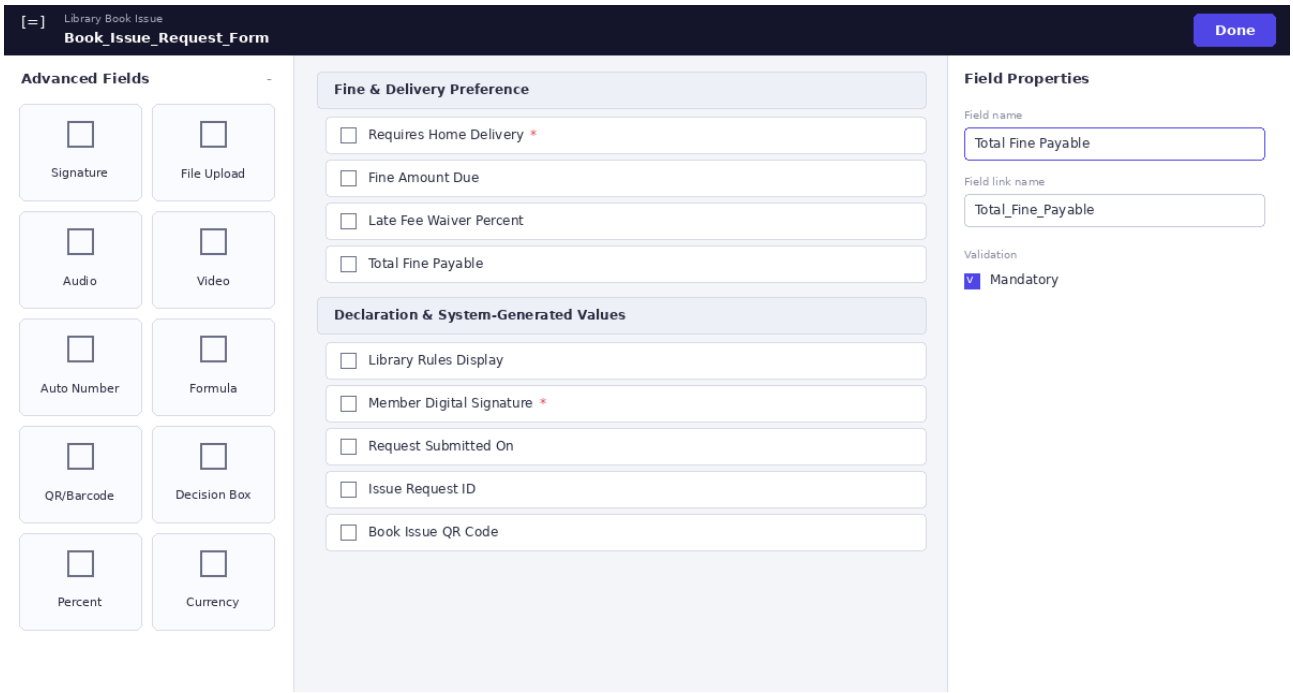


Figure 4B-2: Form Builder - Sections G and H (sample illustration)

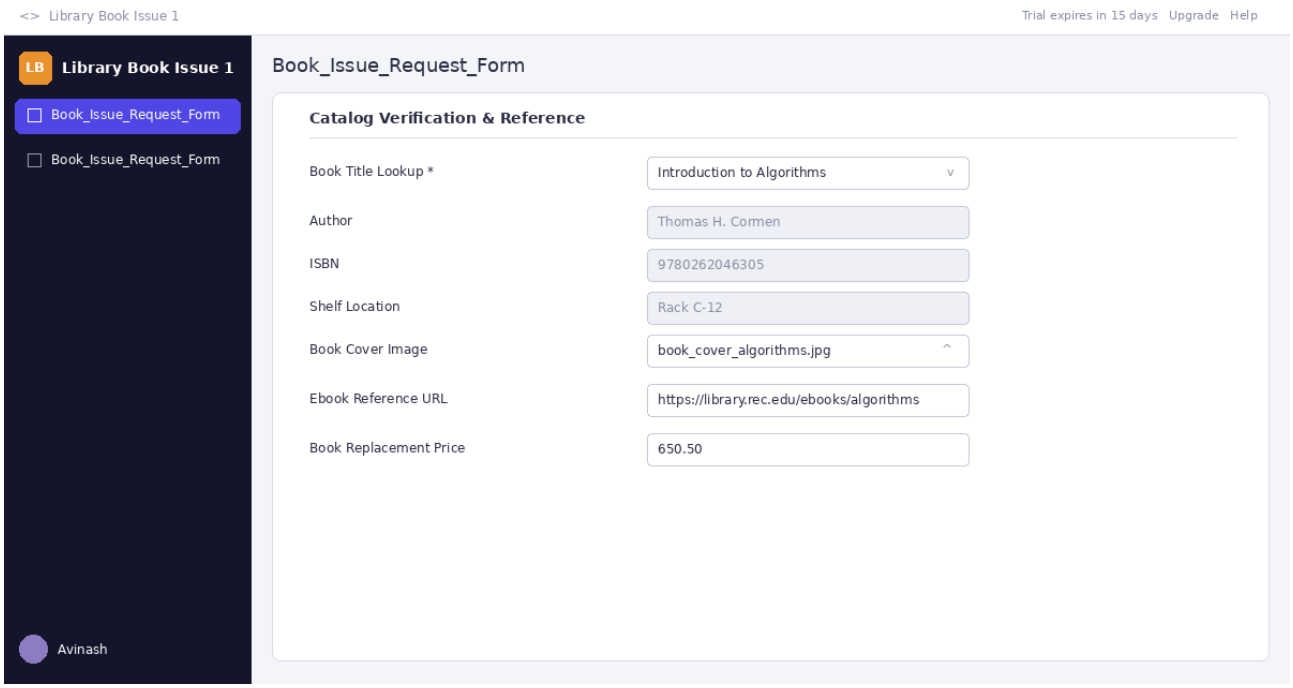


Figure 4B-3: Live form - Book Title Lookup auto-filling author, ISBN, shelf (sample illustration)

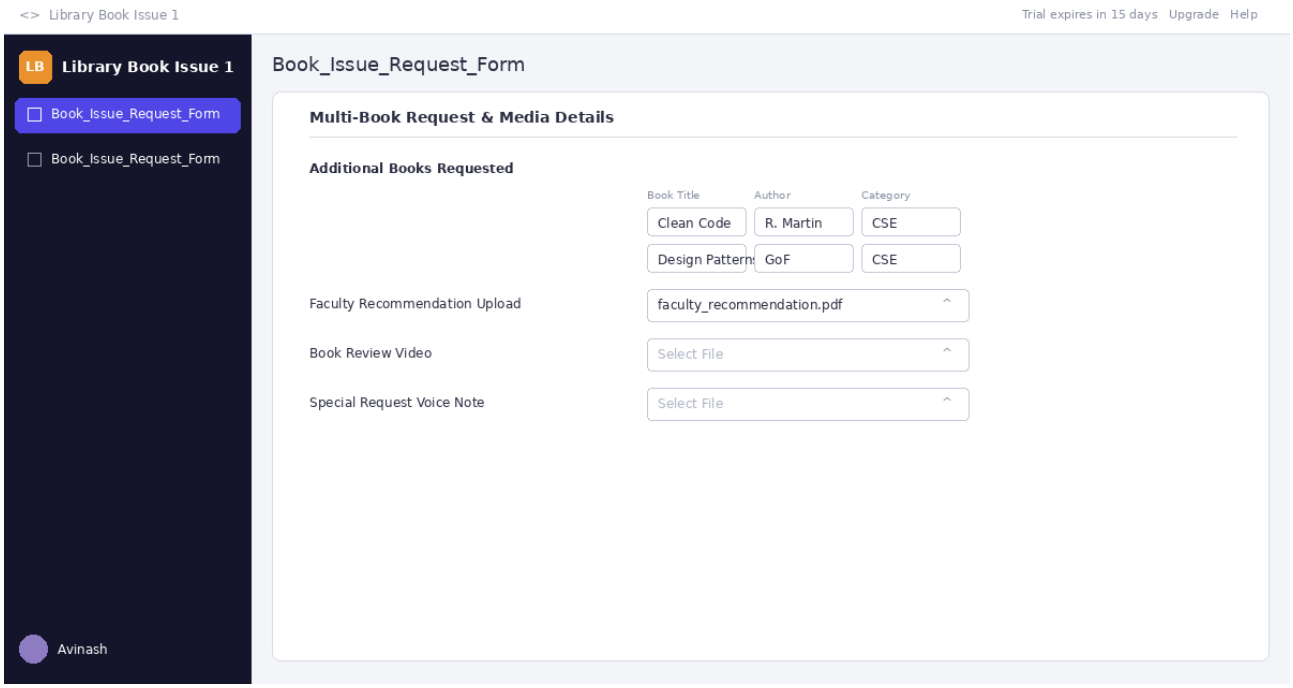


Figure 4B-4: Live form - SubForm rows and media uploads (sample illustration)

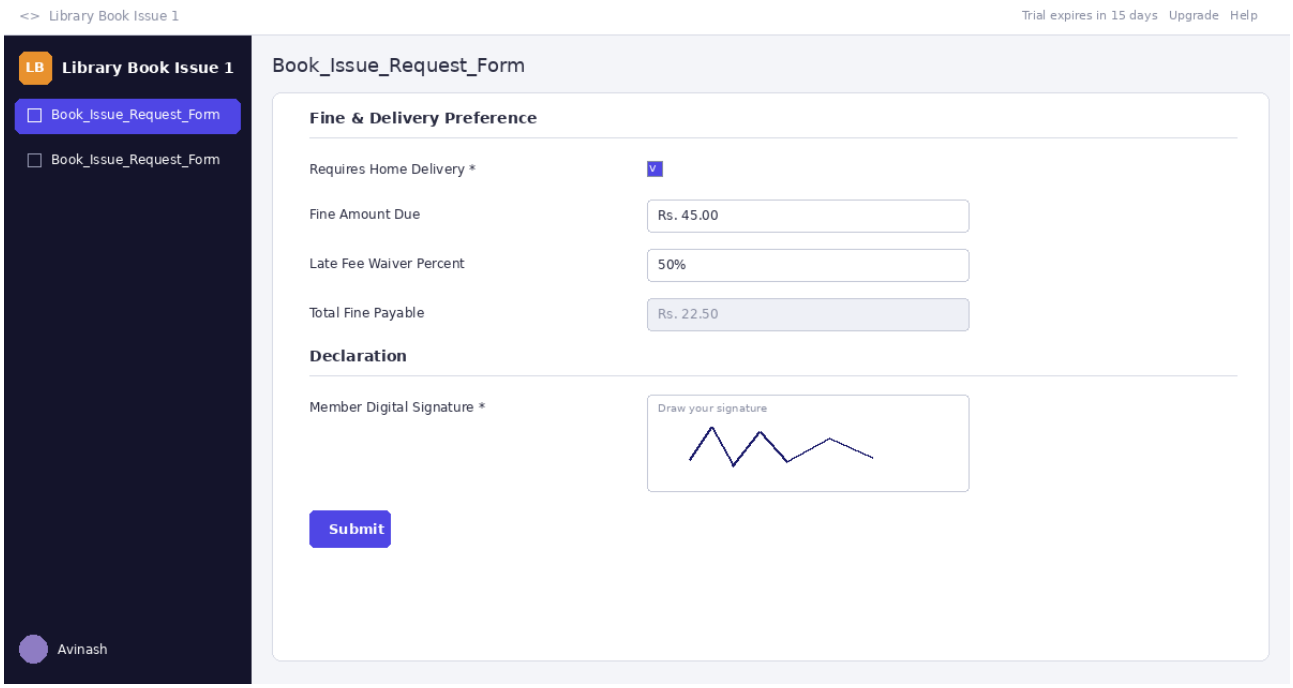


Figure 4B-5: Live form - computed Total Fine Payable and signature (sample illustration)

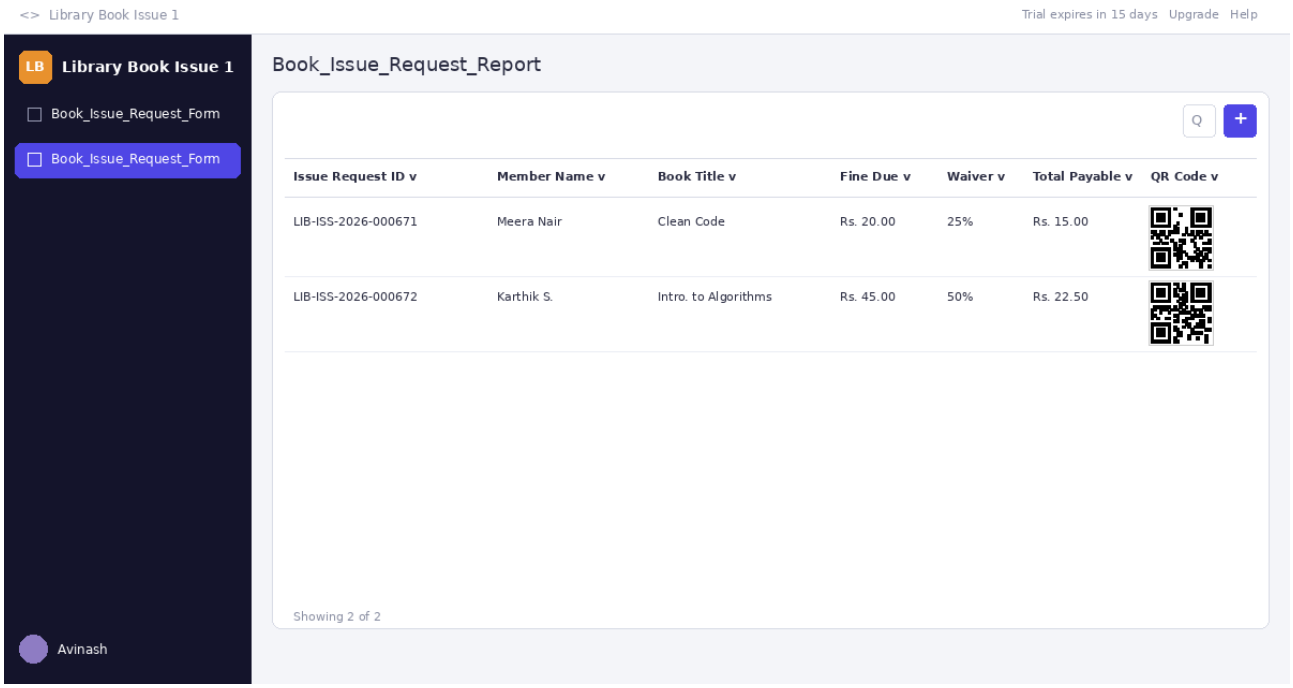


Figure 4B-6: Report with Issue Request ID and QR Code (sample illustration; QR pattern is a placeholder)

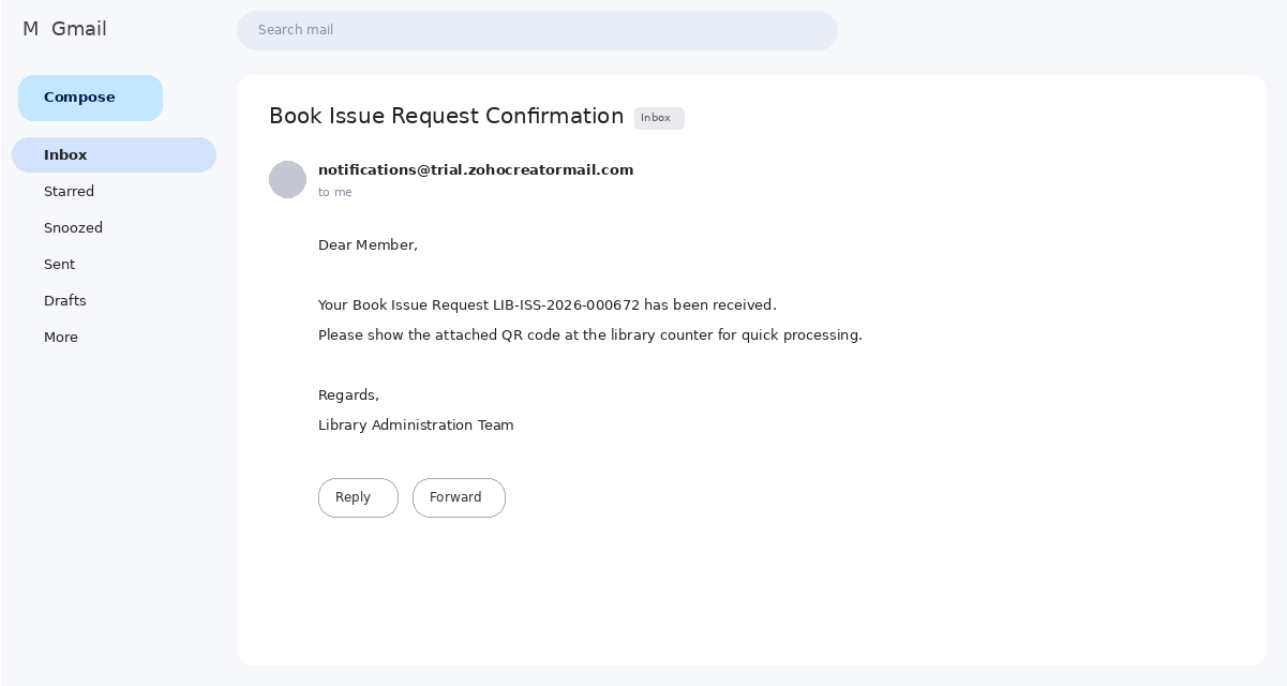


Figure 4B-7: Confirmation email received by the member (sample illustration)

Result

The Library Book Issue Request Form was successfully extended using the required Advanced Fields, validations, and report workflow in Zoho Creator.